

AGILE DEVELOPMENT AND DEVOPS LAB

ASSIGNMENT-5

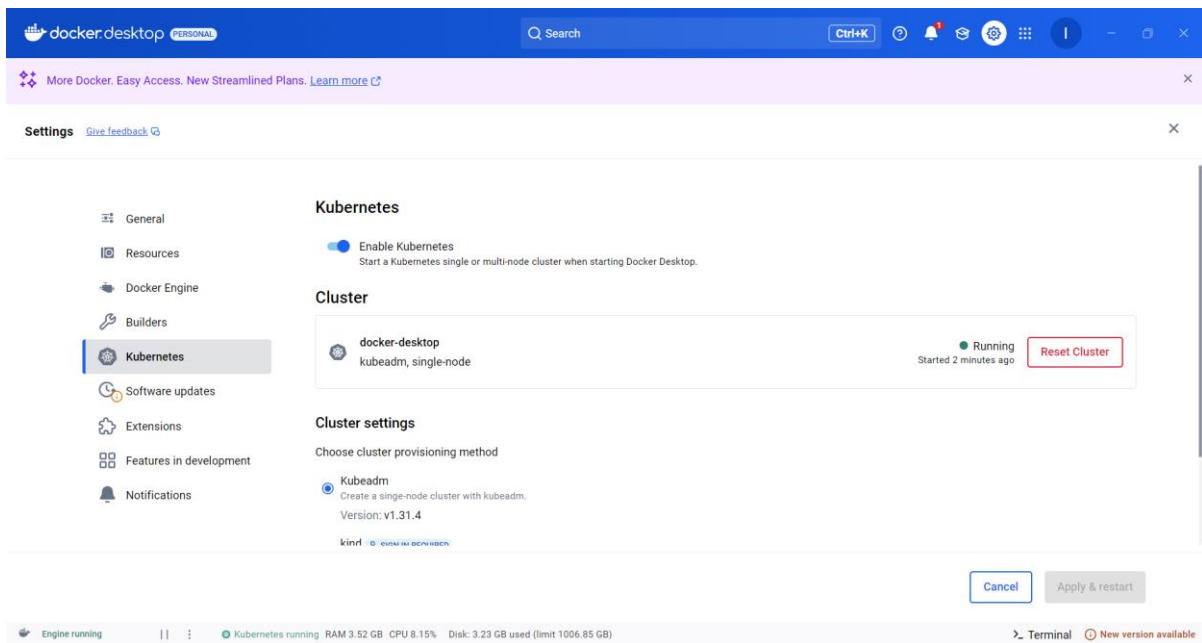
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23MIS0072

I. Jenkin – Docker – Kubernetes (CI-CD Demo)

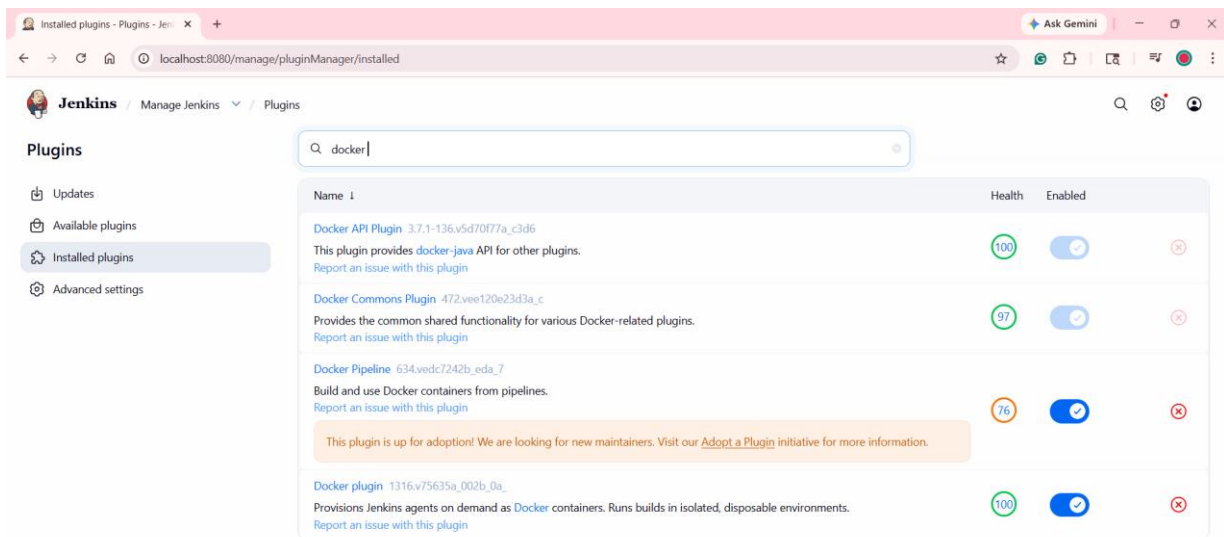
Step1: Install Docker Desktop

Step2: Enable Kubernetes:

Open Docker Desktop Settings -> **Kubernetes** -> Check **Enable Kubernetes**.
This creates a local context named docker-desktop.



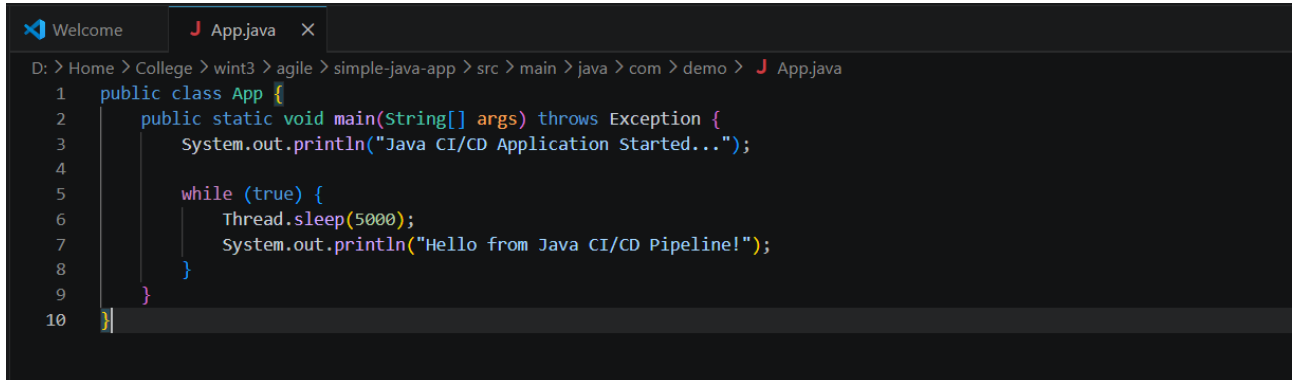
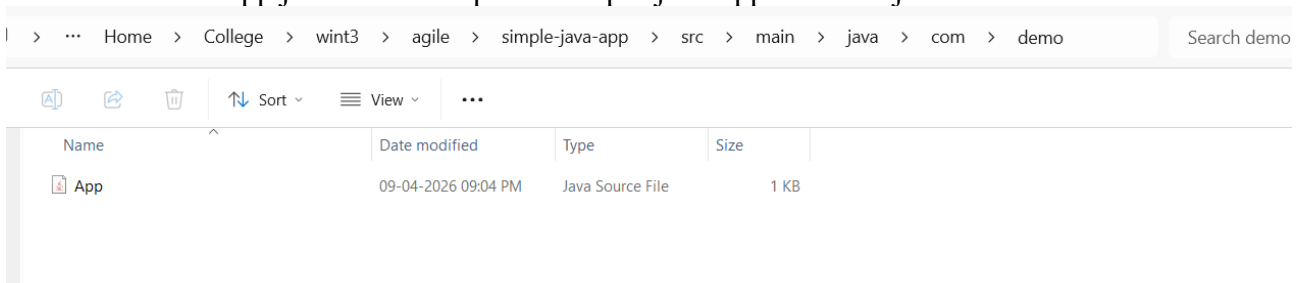
Step 3: In Jenkin, enable two plugins (Docker and Docker Pipeline) and Install them



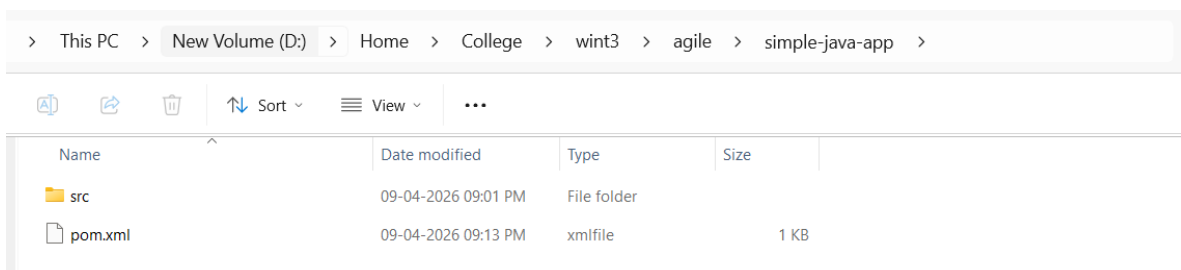
Step 4 : Create Simple Java Application

Create a folder in local machine: simple-java-app

Create Java File App.java in the file path “\simple-java-app\src\main\java\com\demo”



Create Maven File pom.xml in ‘simple-java-app’ folder



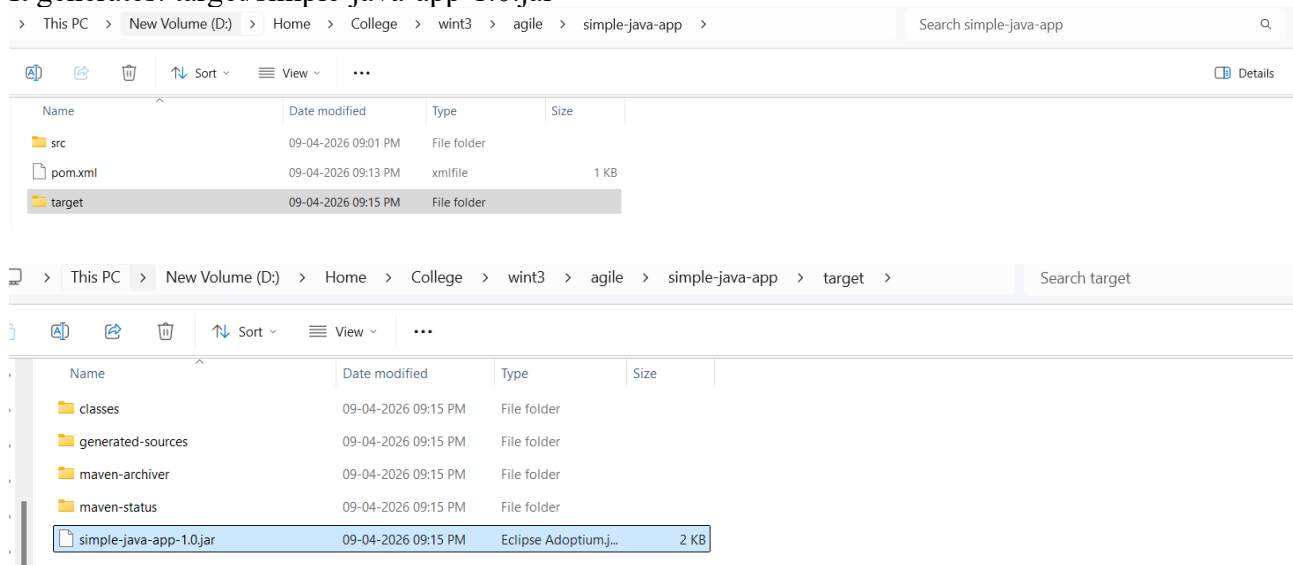
```
Welcome pom.xml
D: > Home > College > wint3 > agile > simple-java-app > pom.xml
1  <project xmlns="http://maven.apache.org/POM/4.0.0"
2      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4      http://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6      <modelVersion>4.0.0</modelVersion>
7
8      <groupId>com.demo</groupId>
9      <artifactId>simple-java-app</artifactId>
10     <version>1.0</version>
11
12     <build>
13         <plugins>
14             <plugin>
15                 <groupId>org.apache.maven.plugins</groupId>
16                 <artifactId>maven-compiler-plugin</artifactId>
17                 <version>3.10.1</version>
18                 <configuration>
19                     <source>17</source>
20                     <target>17</target>
21                 </configuration>
22             </plugin>
23         </plugins>
24     </build>
25 </project>
26
```

Step 5 - Build Locally (Test)

Inside project folder:

mvn clean package

It generates: target/simple-java-app-1.0.jar



```
PS C:\Users\iffah> cd D:\Home\College\wint3\agile\simple-java-app
PS D:\Home\College\wint3\agile\simple-java-app> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.demo:simple-java-app >-----
[INFO] Building simple-java-app 1.0
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.2.0.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-clean-plugin-3.2.0.pom (5.2 kB at 7.7 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.10.1/maven-compiler-plugin-3.10.1.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.10.1/maven-compiler-plugin-3.10.1.pom (10.1 kB at 17.1 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.2.0/maven-jar-plugin-3.2.0.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.2.0/maven-jar-plugin-3.2.0.pom (5.2 kB at 7.7 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.2.0/maven-resources-plugin-3.2.0.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.2.0/maven-resources-plugin-3.2.0.pom (5.2 kB at 7.7 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/2.22.2/maven-surefire-plugin-2.22.2.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/2.22.2/maven-surefire-plugin-2.22.2.pom (5.2 kB at 7.7 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.0.1/maven-install-plugin-3.0.1.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-install-plugin/3.0.1/maven-install-plugin-3.0.1.pom (5.2 kB at 7.7 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-deploy-plugin/3.0.1/maven-deploy-plugin-3.0.1.pom
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-deploy-plugin/3.0.1/maven-deploy-plugin-3.0.1.pom (5.2 kB at 7.7 kB/s)
[INFO] Building jar: D:\Home\College\wint3\agile\simple-java-app\target\simple-java-app-1.0.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 25.596 s
[INFO] Finished at: 2026-04-09T21:15:55+05:30
[INFO]
PS D:\Home\College\wint3\agile\simple-java-app> |
```

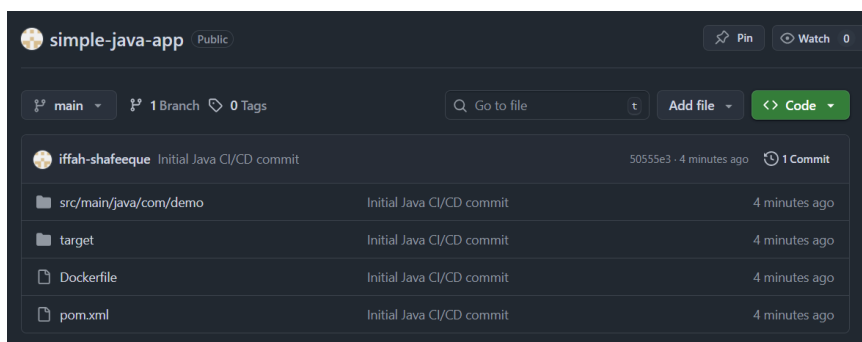
Step 6: Create Dockerfile Dockerfile

```
Welcome Dockerfile X
D: > Home > College > wint3 > agile > simple-java-app > Dockerfile > ...
1 FROM eclipse-temurin:17-jdk
2 WORKDIR /app
3 COPY target/simple-java-app-1.0.jar app.jar
4 CMD ["java", "-jar", "app.jar"]
```

Step 7 : Push Code to GitHub (Before that create a public repository in the Github)

```
git init
git add .
git commit -m "Initial Java CI/CD commit"
git branch -M main
git remote add origin https://github.com/iffah-shafeeqe/simple-java-app
git push -u origin main
```

```
PS D:\Home\College\wint3\agile\simple-java-app> git init
Initialized empty Git repository in D:\Home\College\wint3\agile\simple-java-app\.git/
PS D:\Home\College\wint3\agile\simple-java-app> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time
Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be replaced by CRLF the next time Gi
t touches it
PS D:\Home\College\wint3\agile\simple-java-app> git commit -m "Initial Java CI/CD commit"
[master (root-commit) 50555e3] Initial Java CI/CD commit
8 files changed, 46 insertions(+)
create mode 100644 Dockerfile
create mode 100644 pom.xml
create mode 100644 src/main/java/com/demo/App.java
create mode 100644 target/classes/App.class
create mode 100644 target/maven-archiver/pom.properties
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
create mode 100644 target/simple-java-app-1.0.jar
PS D:\Home\College\wint3\agile\simple-java-app> git branch -M main
PS D:\Home\College\wint3\agile\simple-java-app> git remote add origin https://github.com/iffah-shafeeqe/simple-java-app
PS D:\Home\College\wint3\agile\simple-java-app> git push -u origin main
Enumerating objects: 22, done.
Counting objects: 100% (22/22), done.
Delta compression using up to 8 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (22/22), 3.50 KiB | 358.00 KiB/s, done.
Total 22 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/iffah-shafeeqe/simple-java-app
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS D:\Home\College\wint3\agile\simple-java-app> 
```



Step 8: Create Kubernetes Deployment File *deployment.yaml*

```
! deployment.yaml U X
! deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: java-app
5  spec:
6    replicas: 2
7    selector:
8      matchLabels:
9        app: java-app
10   template:
11     metadata:
12       labels:
13         app: java-app
14     spec:
15       containers:
16         - name: java-app
17           image: iffah11/java-app:latest
18
19   ---
20   apiVersion: v1
21   kind: Service
22   metadata:
23     name: java-service
24   spec:
25     type: NodePort
26     selector:
27       app: java-app
28     ports:
29       - port: 80
30         targetPort: 8080
31         nodePort: 30008
```

After creating it, Push it to GitHub:

```
git add deployment.yaml
git commit -m "Added Kubernetes deployment file"
git push
```

```
PS D:\Home\College\wint3\agile\simple-java-app> git add deployment.yaml
PS D:\Home\College\wint3\agile\simple-java-app> git commit -m "Added Kubernetes deployment file"
[main 3500f77] Added Kubernetes deployment file
1 file changed, 31 insertions(+)
create mode 100644 deployment.yaml
PS D:\Home\College\wint3\agile\simple-java-app> git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 525 bytes | 175.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/iffah-shafeeqe/simple-java-app
50555e3..3500f77  main -> main
PS D:\Home\College\wint3\agile\simple-java-app>
```

Step 9: Configure Jenkins

Run Jenkins

Install Kubernetes CLI plugin

The screenshot shows the Jenkins 'Manage Jenkins' interface, specifically the 'Plugins' tab. On the left, there is a sidebar with navigation options: 'Updates', 'Available plugins', 'Installed plugins' (which is selected), 'Advanced settings', and 'Download progress'. The main area displays a table of installed plugins. A search bar at the top of the table contains the text 'kub'. The table has three columns: 'Name', 'Health', and 'Enabled'. Three plugins are listed:

Name	Health	Enabled
Kubernetes CLI Plugin 1.375.v8001982544ba Configure kubectl for Kubernetes Report an issue with this plugin	100	Yes
Kubernetes Client API Plugin 7.3.1-256.v788a_0b_787114 Kubernetes Client API plugin for use by other Jenkins plugins. Report an issue with this plugin	92	Yes
Kubernetes Credentials Plugin 207.v492f58828b_ed Common classes for Kubernetes credentials Report an issue with this plugin	97	Yes

Step 10: Add DockerHub Credentials

In CI/CD:

1. Jenkins builds Docker image
2. Jenkins pushes image to Docker Hub
3. Kubernetes pulls image from Docker Hub

Without credentials → Push will fail

Steps

Manage Jenkins -> Credentials ->(System)->Global credentials (unrestricted)->Add Credentials

Fill like this:

Field	Value
Kind	Username with password
Scope	Global
Username	your DockerHub username
Password	your DockerHub password
ID	dockerhub-creds
Description	DockerHub Login

Click Create

The screenshot shows the Jenkins 'Credentials' page. The breadcrumb trail is 'Jenkins / Manage Jenkins / Credentials'. The main heading is 'Credentials'. Below this, there is a large light blue box with a person icon and the text 'There are no credentials' and a '+ Add Credentials' button. Below this box, there is a section titled 'Stores scoped to Jenkins'. Under this section, there is a list of stores: 'System' (with a person icon) and 'Global' (with a globe icon). A red arrow points to the 'System' store. Below the 'System' store, there is a sub-section titled 'System'. Under this section, there is a list of domains: 'Global' (with a globe icon). A red arrow points to the 'Global' domain. Below the 'Global' domain, there is a sub-section titled 'Global' with the text '0 credentials - Credentials that should be available everywhere.' and a '+ Add domain' button.

<

Add Username with password

×

Scope ?
Global (Jenkins, nodes, items, all child items, etc) ▼

Username
iffah11

☐ Treat username as secret ?

Password
.....

ID ?
dockerhub-creds

Description ?
DockerHub Login

Create

Jenkins / Manage Jenkins / Credentials / System / Global

Global
Credentials that should be available everywhere.

+ Add Credentials

dockerhub-creds iffah11/*****
DockerHub Login

✎ ⋮

Step 11: Create Jenkins Pipeline Job

Jenkins / All / New Item

New Item

Enter an item name
java-CICD

Select an item type

 Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Pipeline Script:

Script ?

```
1 pipeline {
2     agent any
3
4     environment {
5         DOCKER_IMAGE = "iffah11/java-app:latest"
6     }
7
8     stages {
9         stage('Clone Repository') {
10             steps {
11                 git branch: 'main',
12                     url: 'https://github.com/iffah-shafeeqe/simple-java-app'
13             }
14         }
15     }
16 }
```

```
16 stage('Build with Maven') {
17     steps {
18         bat 'mvn clean package'
19     }
20 }
21
22 stage('Build Docker Image') {
23     steps {
24         bat "docker build -t %DOCKER_IMAGE% ."
25     }
26 }
27
```

```
28 stage('Push to DockerHub') {
29     steps {
30
31         withCredentials([usernamePassword(
32             credentialsId: 'dockerhub-creds',
33             usernameVariable: 'DOCKER_USER',
34             passwordVariable: 'DOCKER_PASS'
35         )]) {
36             bat """
37                 docker login -u %DOCKER_USER% -p %DOCKER_PASS%
38                 docker push %DOCKER_IMAGE%
39                 """
40         }
41     }
42 }
43
44 stage('Deploy to Kubernetes') {
```



```

45      steps {
46          bat 'kubectl apply -f deployment.yaml'
47      }
48  }
49  }
50
51  post {
52      success {
53          echo 'CI/CD Pipeline executed successfully!'
54      }
55      failure {
56          echo 'Pipeline failed. Please check the Console Output for errors.'
57      }
58  }
59 }

```

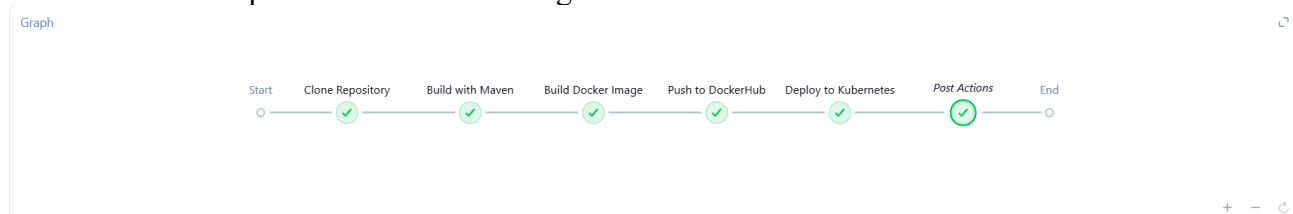
Click → Save → Build Now

```

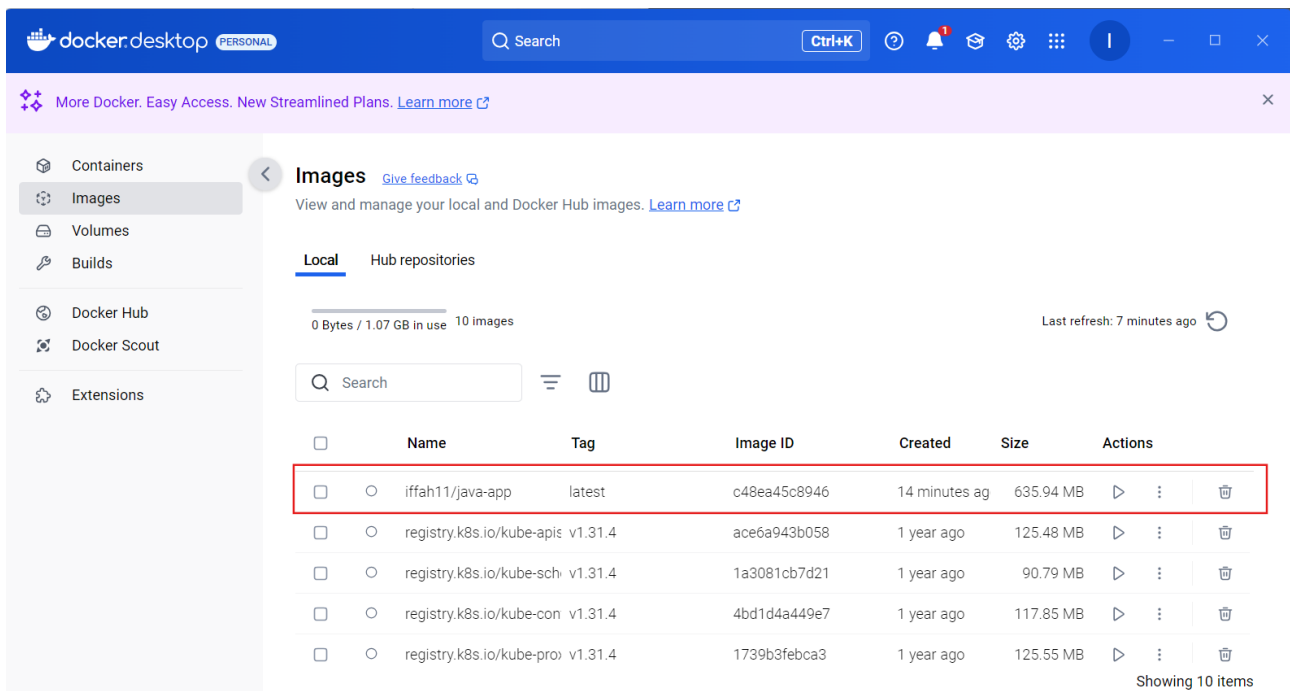
deployment.apps/java-app created
service/java-service created
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

```

Above console output is the success message



Step 12: Verify Docker and Kubernetes Deployment Docker Image



Run this command to verify in command prompt,
 kubectl get svc

```
PS C:\Users\iffah> kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
java-service        NodePort    10.96.78.136  <none>         80:30008/TCP     4m30s
kubernetes           ClusterIP   10.96.0.1     <none>         443/TCP          169m
PS C:\Users\iffah>
```

II. Spring Boot CI/CD Pipeline (GitHub + Jenkins + Docker + Kubernetes)

Step 1: Install Tools: Java 17, Maven, Docker Desktop (enable Kubernetes), Jenkins, and Git.

Step 2: Create Spring Boot Project

pom.xml

```

D: > Home > College > wint3 > agile > SpringProject > SpringProject > pom.xml
1  <project xmlns="http://maven.apache.org/POM/4.0.0"
2      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
4      <modelVersion>4.0.0</modelVersion>
5      <groupId>com.iffah</groupId>
6      <artifactId>app</artifactId>
7      <parent>
8          <groupId>org.springframework.boot</groupId>
9          <artifactId>spring-boot-starter-parent</artifactId>
10         <version>3.2.5</version>
11     </parent>
12     <dependencies>
13         <dependency>
14             <groupId>org.springframework.boot</groupId>
15             <artifactId>spring-boot-starter-web</artifactId>
16         </dependency>
17     </dependencies>
18 </project>

```

App.java

```

pom.xml  App.java  X
D: > Home > College > wint3 > agile > SpringProject > SpringProject > src > main > java > assignment > App.java
1  package assignment;
2
3  import org.springframework.boot.SpringApplication;
4  import org.springframework.boot.autoconfigure.SpringBootApplication;
5
6  @SpringBootApplication
7  public class App {
8      public static void main(String[] args) {
9          SpringApplication.run(App.class, args);
10     }
11 }
12

```

HelloController.java

```

pom.xml  X  App.java  HelloController.java  X
D: > Home > College > wint3 > agile > SpringProject > SpringProject > src > main > java > assignment > HelloController.java
1  package assignment;
2
3  import org.springframework.web.bind.annotation.GetMapping;
4  import org.springframework.web.bind.annotation.RestController;
5
6  @RestController
7  public class HelloController {
8      @GetMapping("/")
9      public String home() { return "App Running"; }
10 }
11

```

Step

3: Run Locally

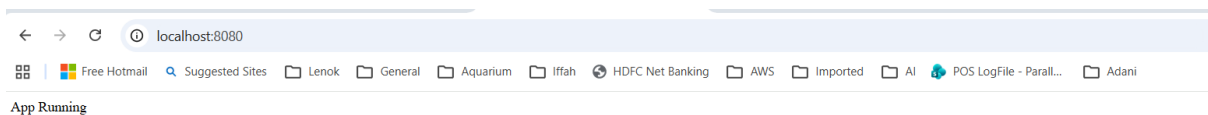
mvn spring-boot:run

```
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ app ---
[INFO] Nothing to compile - all classes are up to date
[INFO] --- maven-resources-plugin:3.3.1:testResources (default-testResources) @ app ---
[INFO] Copying 0 resource from src/test/resources to target/test-classes
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ app ---
[INFO] Nothing to compile - all classes are up to date
[INFO] <<< spring-boot-maven-plugin:3.2.5:run (default-cli) < test-compile @ app <<<
[INFO]
[INFO] --- spring-boot-maven-plugin:3.2.5:run (default-cli) @ app ---
[INFO] Attaching agents: []

  ____  __  __
 / ___/  / /  /
/ /   / /_/ /
/ /___/ __  /
/____/_/ /_/

:: Spring Boot ::
          (v3.2.5)

2026-04-11T22:56:48.717+05:30 INFO 63688 --- [main] assignment.App : Starting App using Java 17.0.12 with PID 63688 (D
: \Home\Iffah\SpringProject\target\classes started by Lenovo in D:\Home\Iffah\SpringProject)
2026-04-11T22:56:48.722+05:30 INFO 63688 --- [main] assignment.App : No active profile set, falling back to 1 default
profile: "default"
2026-04-11T22:56:49.934+05:30 INFO 63688 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-04-11T22:56:49.956+05:30 INFO 63688 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-04-11T22:56:49.957+05:30 INFO 63688 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.20]
2026-04-11T22:56:50.055+05:30 INFO 63688 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-04-11T22:56:50.057+05:30 INFO 63688 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization comple
ted in 1269 ms
2026-04-11T22:56:50.556+05:30 INFO 63688 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context p
ath ''
2026-04-11T22:56:50.578+05:30 INFO 63688 --- [main] assignment.App : Started App in 2.32 seconds (process running for 2.709)
```



Step 4: Docker

Dockerfile

FROM eclipse-temurin:17-jdk

WORKDIR /app

COPY target/app.jar app.jar

ENTRYPOINT ["java", "-jar", "app.jar"]

docker build -t springboot-app .

```
D:\Home\Iffah\SpringProject>docker build -t springboot-app .
[+] Building 48.1s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> transferring dockerfile: 145B
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk
=> [internal] load .dockerignore
=> transferring context: 2B
=> [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:2ff8b91aaa4cb5dc35b4760c37fe064390a486dcd90b361caa72b55d598601f3
=> resolve docker.io/library/eclipse-temurin:17-jdk@sha256:2ff8b91aaa4cb5dc35b4760c37fe064390a486dcd90b361caa72b55d598601f3
=> sha256:ale6e1a167f1d8e27a6ca55c32fa609912f9ff5b0fc88c77413259b59e72ac45 22.97MB / 22.97MB
=> sha256:f00bd8c488bba663bd5f60752b0a1ac92bc39a10729a5a8bf7227b10f6096904 2.28kB / 2.28kB
=> sha256:a7428f4f5c1dad835231adfcff057cad6efc5b2dc256a36871e2e72cb75561e4 145.63MB / 145.63MB
=> sha256:a149c623348d4fa371df99c86694ffdf860874791706585045a923ea4e6224b0d 158B / 158B
=> sha256:689b91d88a0f4086057ec826027b128902ecf2b516be510371c115bc55da19a6 29.73MB / 29.73MB
=> extracting sha256:689b91d88a0f4086057ec826027b128902ecf2b516be510371c115bc55da19a6
=> extracting sha256:ale6e1a167f1d8e27a6ca55c32fa609912f9ff5b0fc88c77413259b59e72ac45
=> extracting sha256:a7428f4f5c1dad835231adfcff057cad6efc5b2dc256a36871e2e72cb75561e4
=> extracting sha256:a149c623348d4fa371df99c86694ffdf860874791706585045a923ea4e6224b0d
=> extracting sha256:f00bd8c488bba663bd5f60752b0a1ac92bc39a10729a5a8bf7227b10f6096904
=> [internal] load build context
=> transferring context: 2.6kB
=> [2/3] WORKDIR /app
=> [3/3] COPY target/app.jar app.jar
=> exporting to image
=> exporting layers
=> exporting manifest sha256:2280f9753ec2e39f57fd8b09cad05a950d973b3cb621a6d07b2873383f8e2cd1
=> exporting config sha256:10f4c5c30c2ea73f92ae602038df712c083fc82818e7439b73dbc57818b22713
=> exporting attestation manifest sha256:3afe873ccbb428b24aca0daf73c0b96b78d7cd8156ef9ae86536da8dff276ef6
=> exporting manifest list sha256:014337f455d5b6fbc8324b5c5d5bec7fbd52e7c62fd00c84af037e3ef00ab4
=> naming to docker.io/library/springboot-app:latest
=> unpacking to docker.io/library/springboot-app:latest

What's next:
View a summary of image vulnerabilities and recommendations → docker scout quickview
```

docker run -p 8080:8080 springboot-app

```
D:\Home\Iffah\SpringProject>docker run -p 8080:8080 springboot-app
no main manifest attribute, in app.jar
```

Step 5: Kubernetes

pom.xml

App.java

HelloController.java

! deployment.yaml

X

D: > Home > College > wint3 > agile > SpringProject > SpringProject > ! deployment.yaml

1

apiVersion: apps/v1

2

kind: Deployment

3

metadata:

4

name: springboot-app

5

spec:

6

replicas: 1

7

template:

8

spec:

9

containers:

10

- name: app

11

image: springboot-app

12

imagePullPolicy: Never

Step 6: GitHub

```
PS D:\Home\College\wint3\agile\SpringProject\SpringProject> git init
Reinitialized existing Git repository in D:/Home/college/wint3/agile/SpringProject/SpringProject/.git/
PS D:\Home\College\wint3\agile\SpringProject\SpringProject> git add .
warning: in the working copy of 'springboot_cicd_guide (3) (2).pdf', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be replaced by CRLF the next time Git touches it
PS D:\Home\College\wint3\agile\SpringProject\SpringProject> git commit -m "init"
On branch master
nothing to commit, working tree clean

PS D:\Home\College\wint3\agile\SpringProject\SpringProject> git remote add origin https://github.com/iffah-shafeeqe/spring
PS D:\Home\College\wint3\agile\SpringProject\SpringProject> git push origin master
Enumerating objects: 43, done.
Counting objects: 100% (43/43), done.
Delta compression using up to 8 threads
Compressing objects: 100% (30/30), done.
Writing objects: 100% (43/43), 9.93 KiB | 1.10 MiB/s, done.
Total 43 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/iffah-shafeeqe/spring
 * [new branch]      master -> master
PS D:\Home\College\wint3\agile\SpringProject\SpringProject> 
```

Step

7: Jenkins Pipeline

Jenkins

/ spring

Configure

General

Triggers

Pipeline

Advanced

Script ?

1~ pipeline {
2 agent any
3
4~ stages {
5~ stage('Clone') {
6~ steps {
7 git 'https://github.com/iffah-shafeeqe/spring'
8 }
9 }
10
11~ stage('Build') {
12~ steps {
13 bat 'mvn clean package'
14 }
15 }
16
17~ stage('Docker') {
18~ steps {
19 bat 'docker build -t springboot-app .'
20 }
21 }
22
23~ stage('Deploy') {
24~ steps {
25 bat 'kubectl apply -f deployment.yaml'
26 }
27 }
28 }
29 }

Save

Apply



Build 18s



Docker Build 22s



Deploy 2s